



National  
Qualifications  
2014

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## **2014 Lifeskills Mathematics Paper 2**

### **National 5**

### **Finalised Marking Instructions**

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## General Marking Principles for National 5 Lifeskills Mathematics

*This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.*

- (a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.
- (b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (c) Credit must be assigned in accordance with the specific assessment guidelines.
- (d) Candidates may use any mathematically correct method to answer questions except in cases where a particular method is specified or excluded.
- (e) Working subsequent to an error must be followed through, with possible credit for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working is easier, candidates lose the opportunity to gain credit.
- (f) Where transcription errors occur, candidates would normally lose the opportunity to gain a processing mark.
- (g) Scored out or erased working which has not been replaced should be marked where still legible. However, if the scored out or erased working has been replaced, only the work which has not been scored out should be judged.
- (h) Unless specifically mentioned in the specific assessment guidelines, do not penalise:
  - Working subsequent to a correct answer
  - Correct working in the wrong part of a question
  - Legitimate variations in solutions
  - Bad form
  - Repeated error within a question

## Detailed Marking Instructions for each question

| Question                                                                                                                                                                                                                                                   |     |  | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                 | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                                                                                                                                                                                                                                                         |     |  | <b>Ans: (£)30, (£)9·30</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: calculate mean</li> <li>•<sup>2</sup> Process: calculate <math>(x - \bar{x})^2</math></li> <li>•<sup>3</sup> Process: substitute into formula</li> <li>•<sup>4</sup> Process: calculate standard deviation</li> </ul> | 4        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(32 + 23...) \div 8 = 30</math></li> <li>•<sup>2</sup> 4, 49, 169, 100, 9, 25, 225, 25</li> <li>•<sup>3</sup> <math>\sqrt{\frac{606}{7}}</math></li> <li>•<sup>4</sup> 9·30</li> </ul>   |
| <b>Notes:</b><br>1. For use of alternative formula; award marks as follows:<br>Mark 2 Process: calculate $\Sigma x$ and $\Sigma x^2$ 240 and 7806<br>Mark 3 Process: substitute into formula<br>Mark 4 Process: calculate standard deviation               |     |  |                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                     |
| 2.                                                                                                                                                                                                                                                         | (a) |  | <b>Ans: Monthly Deal 1 is cheaper</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: find price with Monthly Deal 1</li> <li>•<sup>2</sup> Process: find price with Monthly Deal 2</li> <li>•<sup>3</sup> Communication: state best Deal</li> </ul>                                             | 3        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(279 + 18 + 45 + 9) \times 0.85 = 298.35</math></li> <li>•<sup>2</sup> <math>(18 + 45 + 9) \times 0.35 + 279 = 304.20</math></li> <li>•<sup>3</sup> Monthly Deal 1 is cheaper</li> </ul> |
| <b>Notes:</b><br>1. For “Monthly Deal 1” with no working award 0 marks<br>2. Accept £298/299 for deal 1 and £304/305 for deal 2<br>3. Alternative is by comparing savings.<br>.1 Deal 1 saves £56.25<br>.2 Deal 2 saves £46.80<br>.3 Deal 1 greater saving |     |  |                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                     |

| Question                                                                                                                                                                          |     |      | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                                         | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                   | (b) |      | <b>Ans: £42·19</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: find price for The Red Polka Dot Cycle Shop</li> <li>•<sup>2</sup> Process: find the difference between the price for The Red Polka Dot Cycle Shop and The Yellow Jersey Cycle Shop</li> <li>•<sup>3</sup> Process: calculate total refund</li> </ul> | 3        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(310 + 20 + 50 + 10) \div 3 \times 2 = 260</math></li> <li>•<sup>2</sup> <math>298 \cdot 35 - 260 = 38 \cdot 35</math></li> <li>•<sup>3</sup> <math>38 \cdot 35 \times 1 \cdot 1 = 42 \cdot 19</math></li> </ul> |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>1. Award third mark for £42·18</li> <li>2. The actual cost from deal 1 part a must be used (not a rounded answer)</li> </ol> |     |      |                                                                                                                                                                                                                                                                                                                                        |          |                                                                                                                                                                                                                                                                             |
| 3.                                                                                                                                                                                | (a) |      | <b>Ans: Mark position</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: correct bearing</li> <li>•<sup>2</sup> Process: correct length of line</li> </ul>                                                                                                                                                              | 2        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>065 \pm 2^\circ</math></li> <li>•<sup>2</sup> <math>7 \cdot 6\text{cm} \pm 0 \cdot 2\text{cm}</math></li> </ul>                                                                                                  |
| <b>Notes:</b>                                                                                                                                                                     |     |      |                                                                                                                                                                                                                                                                                                                                        |          |                                                                                                                                                                                                                                                                             |
|                                                                                                                                                                                   | (b) | (i)  | <b>Ans: Mark position</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: bearing from Aberdeen</li> <li>•<sup>2</sup> Strategy: bearing from Ringkobing</li> <li>•<sup>3</sup> Strategy: mark position</li> </ul>                                                                                                      | 3        | <ul style="list-style-type: none"> <li>•<sup>1</sup> Correct bearing of <math>125^\circ \pm 2^\circ</math></li> <li>•<sup>2</sup> Correct bearing of <math>250^\circ \pm 2^\circ</math></li> <li>•<sup>3</sup> Correctly marks position</li> </ul>                          |
|                                                                                                                                                                                   |     | (ii) | <b>Ans: 340km, 200°</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Communication: Distance of fishing vessel from oil rig</li> <li>•<sup>2</sup> Communication: Bearing of fishing vessel from oil rig</li> </ul>                                                                                                            | 2        | <ul style="list-style-type: none"> <li>•<sup>1</sup> Correct distance of <math>340 \pm 10</math></li> <li>•<sup>2</sup> Correct bearing of <math>200^\circ \pm 2^\circ</math></li> </ul>                                                                                    |
| <b>Notes:</b>                                                                                                                                                                     |     |      |                                                                                                                                                                                                                                                                                                                                        |          |                                                                                                                                                                                                                                                                             |

| Question                                                                                                                                            |     |  | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                                                                            | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.                                                                                                                                                  | (a) |  | <b>Ans: £135 000</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: know how to increase by 5%</li> <li>•<sup>2</sup> Strategy: increase for 2 years</li> <li>•<sup>3</sup> Strategy: know how to decrease by 2%</li> <li>•<sup>4</sup> Process: calculate value after 5 years</li> <li>•<sup>5</sup> Communication: round to nearest thousand</li> </ul> | 5        | <ul style="list-style-type: none"> <li>•<sup>1</sup> multiplier of 1.05</li> <li>•<sup>2</sup> <math>130\,000 \times 1.05^2 = (143325)</math></li> <li>•<sup>3</sup> multiplier of 0.98</li> <li>•<sup>4</sup> 134 896.34</li> <li>•<sup>5</sup> 135 000</li> </ul> |
| <b>Notes:</b><br>1. £135 000 without working award 0/5<br>Do not accept £135 000.00                                                                 |     |  |                                                                                                                                                                                                                                                                                                                                                                           |          |                                                                                                                                                                                                                                                                     |
|                                                                                                                                                     | (b) |  | <b>Ans: no value of Saraish's house is about £1000 lower</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: calculate value after 4.5% rise</li> <li>•<sup>2</sup> Communication: compare values</li> </ul>                                                                                                                                                | 2        | <ul style="list-style-type: none"> <li>•<sup>1</sup> 135 850</li> <li>•<sup>2</sup> no value of Saraish's house is lower</li> </ul>                                                                                                                                 |
| <b>Notes:</b> 1. Alternative solution is to compare rises<br>.1 4.5% rise = £5850<br>.2 Saraish's rise is less<br>3 Saraish's rise is 3.8% (< 4.5%) |     |  |                                                                                                                                                                                                                                                                                                                                                                           |          |                                                                                                                                                                                                                                                                     |

| Question                                                                                                    |     |  | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                                                                                                       | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.                                                                                                          | (a) |  | <b>Ans: 9.8 metres</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy/Process: find the hypotenuse</li> <li>•<sup>2</sup> Strategy: know to use correct form of Pythagoras</li> <li>•<sup>3</sup> Process: calculate the length of the wall</li> </ul>                                                                                                                                | 3        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>5 \times 2.8 = 14</math></li> <li>•<sup>2</sup> <math>14^2 - 10^2</math></li> <li>•<sup>3</sup> 9.8</li> </ul>                                                                                                                                     |
| <b>Notes:</b>                                                                                               |     |  |                                                                                                                                                                                                                                                                                                                                                                                                      |          |                                                                                                                                                                                                                                                                                                               |
|                                                                                                             | (b) |  | <b>Ans: £254.15</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: know to calculate area</li> <li>•<sup>2</sup> Process: area of triangle</li> <li>•<sup>3</sup> Process: area of quarter circle</li> <li>•<sup>4</sup> Process: area for turf</li> <li>•<sup>5</sup> Strategy: know how to calculate the number of rolls</li> <li>•<sup>6</sup> Process: calculate cost</li> </ul> | 6        | <ul style="list-style-type: none"> <li>•<sup>1</sup> Rectangle - quarter circle - triangle</li> <li>•<sup>2</sup> 49</li> <li>•<sup>3</sup> 19.6</li> <li>•<sup>4</sup> <math>150 - 49 - 19.6 = 81.4</math></li> <li>•<sup>5</sup> 17</li> <li>•<sup>6</sup> <math>17 \times 14.95 = 254.15</math></li> </ul> |
| <b>Notes:</b><br>1. For mark 6 cost must be stated to 2 decimal places (eg do not accept £342.8 or similar) |     |  |                                                                                                                                                                                                                                                                                                                                                                                                      |          |                                                                                                                                                                                                                                                                                                               |

| Question                                                                                                                                |     |  | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                                                                           | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.                                                                                                                                      | (a) |  | <b>Ans: 0·9s</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: find time difference</li> </ul>                                                                                                                                                                                                                                                           | 1        | <ul style="list-style-type: none"> <li>•<sup>1</sup> 1:50·6 - 1:49·7</li> </ul>                                                                                                                                                                                                              |
| <b>Notes:</b>                                                                                                                           |     |  |                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                                                                                                                                                                                                              |
|                                                                                                                                         | (b) |  | <b>Ans: 179 (km/hr)</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: extract data and substitute</li> <li>•<sup>2</sup> Process: convert time to seconds</li> <li>•<sup>3</sup> Process: calculate speed in km/s</li> <li>•<sup>4</sup> Strategy: know how to convert to km/hr</li> <li>•<sup>5</sup> Communication: round answer correctly</li> </ul> | 5        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>S = 5.543 / 01:51.7</math></li> <li>•<sup>2</sup> 111·7</li> <li>•<sup>3</sup> <math>5.543 / 111.7 = 0.0496...</math></li> <li>•<sup>4</sup> <math>\times 3600</math></li> <li>•<sup>5</sup> 179</li> </ul>                       |
| <b>Notes:</b><br>1. If converted to minutes the evidence would be<br>.2 1·862<br>.3 $5.543 / 1.962 = 2.777$<br>.4 $\times 60$<br>.5 179 |     |  |                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                                                                                                                                                                                                              |
|                                                                                                                                         | (c) |  | <b>Ans: 1 hour 47 minutes 8·8 seconds</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: know to convert time and multiply by 56</li> <li>•<sup>2</sup> Strategy: convert to minutes</li> <li>•<sup>3</sup> Strategy: convert to hours, minutes and seconds</li> <li>•<sup>4</sup> Process: all calculations correct</li> </ul>                          | 4        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>114.8 \times 56 (=6428.8 \text{ secs})</math></li> <li>•<sup>2</sup> <math>\div 60 (107.146...mins)</math></li> <li>•<sup>3</sup> 0·146...mins into seconds (8·8)</li> <li>•<sup>4</sup> 1 hour 47 minutes 8·8 seconds</li> </ul> |
| <b>Notes:</b>                                                                                                                           |     |  |                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                                                                                                                                                                                                              |

| Question      |     |  | Expected Answer(s)<br>Give one mark for each •                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max Mark | Illustrations of evidence for<br>awarding a mark at each •                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.            | (a) |  | <b>Ans: £968·40, £357·48, £741·82</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Process: calculate area of drive in square feet</li> <li>•<sup>2</sup> Process: calculate price for tarmac</li> <li>•<sup>3</sup> Process: calculate how much gravel is needed</li> <li>•<sup>4</sup> Strategy: find best way to buy the gravel</li> <li>•<sup>5</sup> Process: find total cost of using gravel</li> <li>•<sup>6</sup> Strategy: know to calculate minimum number of slabs</li> <li>•<sup>7</sup> Process: calculate number of slabs</li> <li>•<sup>8</sup> Process: calculate amount of hardcore needed</li> <li>•<sup>9</sup> Process: calculate price of slabbed drive</li> </ul> | 9        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>45 \text{ m}^2 \times 10\cdot76 = 484\cdot2 \text{ sq ft}</math></li> <li>•<sup>2</sup> <math>484\cdot2 \times £2 = £968\cdot40</math></li> <li>•<sup>3</sup> <math>45 \times 50 = 2250\text{kg}</math></li> <li>•<sup>4</sup> <math>2 \times 850\text{kg} + 11 \times 50\text{kg}</math></li> <li>•<sup>5</sup> <math>2 \times £125\cdot99 + 11 \times £8\cdot29 + £14\cdot31 = £357\cdot48</math></li> <li>•<sup>6</sup> Evidence</li> <li>•<sup>7</sup> <math>15 \times 15 + 7 \times 7 + 8 = 282</math><br/>Or <math>45 \div 0\cdot16 = 282</math> (rounded up)</li> <li>•<sup>8</sup> <math>45 \text{ m}^2 \times 0\cdot04 \text{ m} = 1\cdot8 \text{ m}^3</math><br/><math>2 \times 2 = 4 \text{ tonnes}</math></li> <li>•<sup>9</sup> <math>282 \times £2\cdot12 + 4 \times £18 + 2 \times £35\cdot99 = £741\cdot82</math></li> </ul> |
| <b>Notes:</b> |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|               | (b) |  | <b>Ans: Choice of surface plus reason</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> Strategy: know to find cost per year for each</li> <li>•<sup>2</sup> Process: calculate the 'cost per year' for each surface type</li> <li>•<sup>3</sup> Communication: state conclusion with valid reason</li> </ul>                                                                                                                                                                                                                                                                                                                                                                            | 3        | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>968\cdot40 \div 30, 357\cdot48 \div 10, 741\cdot82 \div 25</math></li> <li>•<sup>2</sup> Tarmac costs £32·28 per year<br/>Gravel costs £35·75 per year<br/>Slabs cost £29·67 per year</li> <li>•<sup>3</sup> Slabs cheapest per year, or gravel cheaper initially etc</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Notes:</b> |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

[END OF MARKING INSTRUCTIONS]